Solutions

Maths

Q1. Let $\mathcal{F}$ be the family of curves given by $x^2+y^2=C\exp\left(\frac{y}{x}\right)$ where $C>0$ is an arbitrary constant. Let $\mathcal{C}$ be the orthogonal trajectory of the family $\mathcal{F}$ that passes through the point $(1,0)$. If (r,θ) represents the polar coordinates of any point on $\mathcal{C}$ with $r>0$ and $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, then the distance of the point on $\mathcal{C}$ lying on the line $y=x$ $(x>0)$ from the origin is equal to:

(A) $e^{-\frac{\pi+2}{4}}$ (B) $e^{-\frac{\pi+1}{4}}$ (C) $e^{-\frac{\pi+4}{4}}$ (D) $e^{-\frac{\pi+2}{2}}$

Ans. A

Sol. To find the differential equation of the family $\mathcal{F}$, we use polar coordinates $x=r\cos\theta$ and $y=r\sin\theta$.

Substituting $x^2+y^2=r^2$ and $\frac{y}{x}=\tan\theta$ into the equation of $\mathcal{F}$: $r^2=Ce^{\tan\theta}$

Taking natural logarithm on both sides: $2\ln r=\ln C+\tan\theta$

Differentiating both sides with respect to θ to eliminate C: $\frac{2}{r}\frac{dr}{d\theta}=\sec^2\theta \implies \frac{dr}{d\theta}=\frac{r}{2}\sec^2\theta$

For the family of orthogonal trajectories $\mathcal{C}$, we replace $\frac{dr}{d\theta}$ by $-r^2\frac{d\theta}{dr}$: $-r^2\frac{d\theta}{dr}=\frac{r}{2}\sec^2\theta$

Separating the variables r and θ: $\frac{dr}{r}=-2\frac{d\theta}{\sec^2\theta}=-2\cos^2\theta d\theta=-(1+\cos2\theta)d\theta$

Integrating both sides: $\int\frac{dr}{r}=-\int(1+\cos2\theta)d\theta \implies \ln r=-\theta-\frac{1}{2}\sin2\theta+K$

Since the trajectory passes through $(x,y)=(1,0)$, in polar coordinates $(r,\theta)=(1,0)$:

$\ln1=-0-\frac{1}{2}\sin0+K \implies K=0$

Thus, the equation of the orthogonal trajectory $\mathcal{C}$ is: $\ln r=-\theta-\frac{1}{2}\sin2\theta \implies r(\theta)=\exp\left(-\theta-\frac{1}{2}\sin2\theta\right)$

For the point lying on the line $y=x$ with $x>0$, we have $\tan\theta=1 \implies \theta=\frac{\pi}{4}$.

The distance of this point from the origin is $r\left(\frac{\pi}{4}\right)$: $r\left(\frac{\pi}{4}\right)=\exp\left(-\frac{\pi}{4}-\frac{1}{2}\sin\left(\frac{\pi}{2}\right)\right)$

$=\exp\left(-\frac{\pi}{4}-\frac{1}{2}\right)=e^{-\frac{\pi+2}{4}}$

Q2. Let $f:(0,\infty)\to\mathbb{R}$ be a function defined by

$f(x)=\int\frac{2x^{12}+5x^9}{(x^5+x^3+1)^3}\,dx$ such that $f(1)=\frac{1}{18}$.

If $a = \lim_{x\to\infty} f(x)$, consider the parabola $y^2 = 4ax$. Let A be a point on its directrix having y-coordinate equal to 2. If the tangents drawn from A to the parabola touch it at points P and Q, then the length of the chord PQ is:

(A) 5 (B) 10 (C) 12 (D) 15

Ans. B

Sol. Dividing the numerator and denominator of the integrand by x^{15}:

$$f(x) = \int \frac{2x^{-3} + 5x^{-6}}{\left(1 + x^{-2} + x^{-5}\right)^3}\, dx$$

Let $u = 1 + x^{-2} + x^{-5} \implies du = -\left(2x^{-3} + 5x^{-6}\right)dx$.

$$f(x) = \int \frac{-du}{u^3} = \frac{1}{2u^2} + C = \frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$$

Given $f(1) = \frac{1}{18}$. $\implies \frac{1}{18} + C = \frac{1}{18} \implies C = 0$

Thus, $f(x) = \frac{x^{10}}{2(x^5 + x^3 + 1)^2}$.

Now, $a = \lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x^{10}}{2x^{10}\left(1 + x^{-2} + x^{-5}\right)^2} = \frac{1}{2}$.

The given parabola is $y^2 = 2x$ (since $a = 1/2$). The directrix of $y^2 = 2x$ is $x = -a = -\frac{1}{2}$.

Thus, the point A is $\left(-\frac{1}{2}, 2\right)$.

The chord of contact PQ for point $A\left(-\frac{1}{2}, 2\right)$ with respect to $y^2 = 2x$ is given by $T = 0$:

$$y(2) = 1 \cdot \left(x - \frac{1}{2}\right) \implies 2x - 4y - 1 = 0 \implies x = 2y + \frac{1}{2}$$

Substituting $x = 2y + \frac{1}{2}$ into $y^2 = 2x$:

$$y^2 = 2\left(2y + \frac{1}{2}\right) \implies y^2 - 4y - 1 = 0$$

Let y_1, y_2 be the roots.

Then $y_1 + y_2 = 4$ and $y_1 y_2 = -1$.

$$|y_1 - y_2| = \sqrt{(y_1 + y_2)^2 - 4y_1y_2} = \sqrt{16 + 4} = \sqrt{20}$$

Since $x_1 - x_2 = 2(y_1 - y_2)$, the length of chord PQ is: $PQ = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

$$= \sqrt{5(y_1 - y_2)^2} = \sqrt{5 \times 20} = 10$$

Q3. Let $f : (0, 1] \to \mathbb{R}$ be a non-constant differentiable function such that $f(1) = 1$ and $f\left(\frac{1}{2}\right) = 1 + \frac{\sqrt{3}}{2}$. For any point $P(x, f(x))$ on the curve $y = f(x)$, the perpendicular distance from the origin $(0, 0)$ to the normal to the curve at P is equal to x. If $g(x)$ denotes the x-intercept of the tangent to the curve $y = f(x)$ at the point $P(x, f(x))$, then the value of the definite integral $\int_{1/2}^{1} g(x)\, dx$ is equal to:

(A) $2\ln(\sqrt{3} + 1) - \frac{\sqrt{3}}{2}$ (B) $\ln(\sqrt{3} + 1) - \frac{\sqrt{3}}{4}$ (C) $2\ln(\sqrt{3} - 1) + \frac{\sqrt{3}}{2}$

(D) $\ln(2+\sqrt{3})-\sqrt{3}$

Ans. A

Sol. Let $y=f(x)$. The equation of the normal to the curve at $P(x,y)$ is: $Y-y=-\dfrac{1}{y'}(X-x)$

$\implies X+y'Y-(x+yy')=0$

The perpendicular distance from $(0,0)$ to this normal is given as x: $\dfrac{|x+yy'|}{\sqrt{1+(y')^2}}=x$

$\implies (x+yy')^2=x^2(1+(y')^2)$

Expanding and simplifying: $x^2+2xyy'+y^2(y')^2=x^2+x^2(y')^2$

$\implies y'\left[(x^2-y^2)y'-2xy\right]=0$

Since $f(x)$ is non-constant, $y'\neq 0$. Thus: $\dfrac{dy}{dx}=\dfrac{2xy}{x^2-y^2}$

Solving this homogeneous differential equation by substituting $y=vx$:

$v+x\dfrac{dv}{dx}=\dfrac{2v}{1-v^2}\implies x\dfrac{dv}{dx}=\dfrac{v(1+v^2)}{1-v^2}$

$\displaystyle\int\left(\frac{1}{v}-\frac{2v}{1+v^2}\right)dv=\int\frac{dx}{x}\implies \ln\left|\frac{v}{1+v^2}\right|=\ln|x|+C$

Resubstituting $v=y/x$: $\dfrac{y}{x^2+y^2}=c\implies x^2+y^2-2ky=0$

Using $f(1)=1\implies 1+1-2k=0\implies k=1$.

Thus, $x^2+(y-1)^2=1$. Since $f(1/2)=1+\dfrac{\sqrt{3}}{2}$, we take the upper branch: $y=f(x)=1+\sqrt{1-x^2}$

The slope of the tangent at $P(x,y)$ is $y'=-\dfrac{x}{\sqrt{1-x^2}}$.

The equation of the tangent is $Y-y=y'(X-x)$

.Setting $Y=0$, the x-intercept $g(x)$ is:

$g(x)=x-\dfrac{y}{y'}=x-\dfrac{1+\sqrt{1-x^2}}{-\frac{x}{\sqrt{1-x^2}}}=\dfrac{1+\sqrt{1-x^2}}{x}$

Now, we evaluate $I=\displaystyle\int_{1/2}^{1}\frac{1+\sqrt{1-x^2}}{x}\,dx$.

Note that $\dfrac{d}{dx}\left[\ln\left(1-\sqrt{1-x^2}\right)+\sqrt{1-x^2}\right]=\dfrac{1+\sqrt{1-x^2}}{x}$.

Therefore:

$I=\left[\ln\left(1-\sqrt{1-x^2}\right)+\sqrt{1-x^2}\right]_{1/2}^{1}$

$I=(0+0)-\left(\ln\left(1-\dfrac{\sqrt{3}}{2}\right)+\dfrac{\sqrt{3}}{2}\right)=-\ln\left(\dfrac{2-\sqrt{3}}{2}\right)-\dfrac{\sqrt{3}}{2}$

$I=\ln\left(\dfrac{2}{2-\sqrt{3}}\right)-\dfrac{\sqrt{3}}{2}=\ln(4+2\sqrt{3})-\dfrac{\sqrt{3}}{2}=\ln(\sqrt{3}+1)^2-\dfrac{\sqrt{3}}{2}=2\ln(\sqrt{3}+1)-\dfrac{\sqrt{3}}{2}$

Q4. Let $y(x)$ be a differentiable function defined for $x>0$ satisfying the differential equation $(x+x^2e^x)\dfrac{dy}{dx}+(1+2xe^x+x^2e^x)y=x^2e^x$ If $y(1)=\dfrac{e}{1+e}$, then the value of $y(2)$ is equal to

(A) $\dfrac{e^2}{1+2e^2}$ (B) $\dfrac{e^2}{2+e^2}$ (C) $\dfrac{2e^2}{1+e^2}$ (D) $\dfrac{e^2}{1+e^2}$

Ans. A

Sol. The given differential equation can be rewritten as: $\frac{dy}{dx} + \left(\frac{1 + 2xe^x + x^2e^x}{x(1 + xe^x)} \right) y = \frac{x^2e^x}{x(1 + xe^x)}$

This is a linear differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$,

where $P(x) = \frac{(1 + xe^x) + xe^x(1 + x)}{x(1 + xe^x)} = \frac{1}{x} + \frac{e^x(1 + x)}{1 + xe^x}$

The integrating factor (I.F.) is: I.F. $= e^{\int P(x)\,dx} = e^{\ln x + \ln(1 + xe^x)} = x(1 + xe^x)$ Multiplying by I.F.,

the differential equation becomes an exact differential:

$\frac{d}{dx}[y \cdot x(1 + xe^x)] = x(1 + xe^x) \cdot \frac{xe^x}{1 + xe^x} = x^2e^x$ Alternatively,

by grouping terms as exact differentials: $d(xy) + d(x^2ye^x) = x^2e^x dx$

$\implies d\left[xy(1 + xe^x)\right] = x^2e^x dx$

Integrating both sides with respect to x: $xy(1 + xe^x) = \int x^2e^x\,dx = (x^2 - 2x + 2)e^x + C$

Using the initial condition $y(1) = \frac{e}{1 + e}$: $1 \cdot \left(\frac{e}{1 + e} \right)(1 + e) = (1 - 2 + 2)e + C$

$\implies e = e + C \implies C = 0$

Thus, the explicit function is: $y(x) = \frac{(x^2 - 2x + 2)e^x}{x(1 + xe^x)}$

Substituting $x = 2$: $y(2) = \frac{(4 - 4 + 2)e^2}{2(1 + 2e^2)} = \frac{e^2}{1 + 2e^2}$

Q5. Consider the parabola $\mathcal{P} : y^2 = 4x$. Let $P(\alpha, \beta)$ be a point from which three real and distinct normals are drawn to $\mathcal{P}$, intersecting $\mathcal{P}$ at the points T_1, T_2, and T_3. Let S be the circle passing through T_1, T_2, T_3 and the origin $O(0, 0)$.

Which of the following statement(s) is/are **CORRECT**?

(A) For any $\alpha \in (2, 8)$, every point $P(\alpha, \beta)$ from which three distinct normals can be drawn to $\mathcal{P}$ lies strictly inside the parabola $\mathcal{P}$.

(B) The centroid G of $\triangle T_1T_2T_3$ lies strictly inside the parabola $\mathcal{P}$ for all permissible positions of $P(\alpha, \beta)$.

(C) The center of the circle S lies strictly outside the parabola $\mathcal{P}$ for all $\alpha \in (2, 8)$.

(D) The orthocenter H of $\triangle T_1T_2T_3$ lies strictly outside the parabola $\mathcal{P}$ for all $\alpha \in (2, 6)$.

Ans. A, B, D

Sol. The equation of a normal to $y^2 = 4x$ is $y = mx - 2m - m^3$.

Since it passes through $P(\alpha, \beta)$, we have $m^3 + (2 - \alpha)m + \beta = 0$. Let m_1, m_2, m_3 be the roots.

Then $m_1 + m_2 + m_3 = 0$, $m_1m_2 + m_2m_3 + m_3m_1 = 2 - \alpha$, $m_1m_2m_3 = -\beta$.

For three real and distinct normals, the discriminant must be positive: $27\beta^2 < 4(\alpha - 2)^3$,which implies $\alpha > 2$.

Option (A): A point $P(\alpha, \beta)$ is inside $y^2 = 4x$ if $\beta^2 - 4\alpha < 0$.

Since $27\beta^2 < 4(\alpha - 2)^3$, we have $\beta^2 < \frac{4}{27}(\alpha - 2)^3$.

Let $f(\alpha) = 4\alpha - \frac{4}{27}(\alpha - 2)^3$.

Then $f'(\alpha) = 4 - \frac{4}{9}(\alpha - 2)^2 = \frac{4}{9}(9 - (\alpha - 2)^2)$.

For $\alpha \in (2, 8)$, $f'(\alpha) > 0$ on $(2, 5)$ and $f'(\alpha) < 0$ on $(5, 8)$.

Since $f(2) = 8 > 0$ and $f(8) = 0$, $f(\alpha) > 0$ for $\alpha \in (2, 8)$.

Thus, $\beta^2 < \frac{4}{27}(\alpha - 2)^3 < 4\alpha$, so $\beta^2 - 4\alpha < 0$. $P(\alpha, \beta)$ is strictly inside $\mathcal{P}$.

Option (B): The feet of normals are $T_i(m_i^2, -2m_i)$.

The centroid $G = \left(\frac{\sum m_i^2}{3}, \frac{-2\sum m_i}{3} \right) = \left(\frac{-2(2 - \alpha)}{3}, 0 \right) = \left(\frac{2(\alpha - 2)}{3}, 0 \right)$.

For G to be inside $\mathcal{P}$, $0^2 - 4\left(\frac{2(\alpha - 2)}{3} \right) < 0$, which is $-\frac{8(\alpha - 2)}{3} < 0$.

This is true for $\alpha > 2$. Thus, G is strictly inside $\mathcal{P}$.

The circle S through T_1, T_2, T_3 and $(0, 0)$ has equation $x^2 + y^2 + 2gx + 2fy = 0$.

Substituting $x = t^2, y = -2t$ gives $t^4 + (4 + 2g)t^2 - 4ft = 0$, so $t^3 + (4 + 2g)t - 4f = 0$.

Comparing with $t^3 + (2 - \alpha)t + \beta = 0$, we get $4 + 2g = 2 - \alpha$

$\implies g = -\frac{\alpha + 2}{2}$ and $-4f = \beta \implies f = -\frac{\beta}{4}$

The center is $C = (-g, -f) = \left(\frac{\alpha + 2}{2}, \frac{\beta}{4} \right)$.

For C to be inside $\mathcal{P}$, $\left(\frac{\beta}{4} \right)^2 - 4\left(\frac{\alpha + 2}{2} \right) < 0$. $\implies \frac{\beta^2}{16} - 2(\alpha + 2) < 0$

Using $\beta^2 < \frac{4}{27}(\alpha - 2)^3$, we get $\frac{(\alpha - 2)^3}{108} - 2\alpha - 4 < 0$.

Let $g(\alpha) = \frac{(\alpha - 2)^3}{108} - 2\alpha - 4$. $g'(\alpha) = \frac{(\alpha - 2)^2}{36} - 2 < 0$ for $\alpha \in (2, 8)$.

Since $g(2) = -8 < 0$, $g(\alpha) < -8 < 0$ for $\alpha \in (2, 8)$. Thus, C is strictly inside $\mathcal{P}$

Option (D): The orthocenter $H = \left(\alpha - 6, -\frac{\beta}{2} \right)$.

For H to be outside $\mathcal{P}$, $\left(-\frac{\beta}{2} \right)^2 - 4(\alpha - 6) > 0$. $\implies \frac{\beta^2}{4} + 4(6 - \alpha) > 0$

Since $\beta^2 < \frac{4}{27}(\alpha - 2)^3$, we need to check if $\frac{(\alpha - 2)^3}{27} + 4(6 - \alpha) > 0$.

Let $h(\alpha) = \frac{(\alpha - 2)^3}{27} + 24 - 4\alpha$. $h'(\alpha) = \frac{(\alpha - 2)^2}{9} - 4$.

For $\alpha \in (2, 6)$, $h'(\alpha) < 0$. $h(2) = 24 - 8 = 16 > 0$. $h(6) = 0 + 24 - 24 = 0$.

Since $h(\alpha)$ is decreasing on $(2, 6)$ and $h(6) = 0$, $h(\alpha) > 0$ for $\alpha \in (2, 6)$.

Thus, H is strictly outside $\mathcal{P}$.

Q6. Let $f : [-2, 2] \to \mathbb{R}$ be a function defined as:

$$f(x) = \lim_{n\to\infty} \frac{x^{2n} \sin\left(\frac{\pi x}{2}\right) + x^2}{x^{2n} + 1}$$

Which of the following statement(s) is/are **CORRECT**?

(A) $f(x)$ is continuous on $[-2, 2] \setminus \{-1\}$.

(B) The total area bounded by the curve $y = f(x)$, the x-axis, and the vertical lines $x = -2$ and $x = 2$ is $\frac{4}{\pi} + \frac{1}{3}$.

(C) $\int_{-2}^{2} f(x)\, dx = \frac{2}{3}$.

(D) The area of the region bounded by $y = f(x)$, the line $y = 1$, and the y-axis for $x \in [0, 2]$ is $\frac{5}{3} - \frac{2}{\pi}$.

Ans. A, C

Sol. Evaluating the limit definition of $f(x)$ for different intervals of x:

1. For $|x| < 1 \implies x^{2n} \to 0$ as $n \to \infty$, so $f(x) = x^2$.

2. For $|x| > 1 \implies x^{2n} \to \infty$ as $n \to \infty$, so $f(x) = \sin\left(\frac{\pi x}{2}\right)$.

3. For $x = 1$, $f(1) = \frac{1 \cdot \sin(\pi/2) + 1}{1 + 1} = 1$.

4. For $x = -1$, $f(-1) = \frac{1 \cdot \sin(-\pi/2) + 1}{1 + 1} = 0$.

Thus, $f(x)$ is defined piecewise as: $f(x) = \begin{cases} \sin\left(\frac{\pi x}{2}\right), & x \in [-2, -1) \cup (1, 2] \\ x^2, & x \in (-1, 1) \\ 0, & x = -1 \\ 1, & x = 1 \end{cases}$

Checking Continuity:- At $x = 1$: $\lim\limits_{x\to1^-} f(x) = 1$, $\lim\limits_{x\to1^+} f(x) = \sin\left(\frac{\pi}{2}\right) = 1$, and $f(1) = 1$.

Hence, continuous at $x = 1$. - At $x = -1$: $\lim\limits_{x\to-1^-} f(x) = \sin\left(-\frac{\pi}{2}\right) = -1$, whereas $\lim\limits_{x\to-1^+} f(x) = 1$.

Discontinuous at $x = -1$. Hence, $f(x)$ is continuous on $[-2, 2] \setminus \{-1\}$. Option (A) is correct.

Evaluating the Definite Integral:

$$\int_{-2}^{2} f(x)\, dx = \int_{-2}^{-1} \sin\left(\frac{\pi x}{2}\right) dx + \int_{-1}^{1} x^2\, dx + \int_{1}^{2} \sin\left(\frac{\pi x}{2}\right) dx$$

$$\int_{-2}^{-1} \sin\left(\frac{\pi x}{2}\right) dx = \left[-\frac{2}{\pi}\cos\left(\frac{\pi x}{2}\right)\right]_{-2}^{-1} = -\frac{2}{\pi}$$

$$\int_{-1}^{1} x^2\, dx = \frac{2}{3}$$

$$\int_{1}^{2} \sin\left(\frac{\pi x}{2}\right) dx = \left[-\frac{2}{\pi}\cos\left(\frac{\pi x}{2}\right)\right]_{1}^{2} = \frac{2}{\pi} \implies \int_{-2}^{2} f(x)\, dx = -\frac{2}{\pi} + \frac{2}{3} + \frac{2}{\pi} = \frac{2}{3}$$

Calculating Total Bounded Area A :

Notice that $f(x) \leq 0$ on $[-2, -1]$ and $f(x) \geq 0$ on $[-1, 2]$.

$$A = \int_{-2}^{2} |f(x)|\, dx = \int_{-2}^{-1} \left(-\sin\left(\frac{\pi x}{2}\right)\right) dx + \int_{-1}^{1} x^2\, dx + \int_{1}^{2} \sin\left(\frac{\pi x}{2}\right) dx$$

$$A = \frac{2}{\pi} + \frac{2}{3} + \frac{2}{\pi} = \frac{4}{\pi} + \frac{2}{3}$$

Area bounded by $y = f(x)$, $y = 1$, and y-axis for $x \in [0, 2]$:

Since $f(x) \leq 1$ for all $x \in [0, 2]$, the area is:

$$\text{Area} = \int_{0}^{2} (1 - f(x))\, dx = \int_{0}^{2} 1\, dx - \int_{0}^{1} x^2\, dx - \int_{1}^{2} \sin\left(\frac{\pi x}{2}\right) dx$$

$$\text{Area} = 2 - \frac{1}{3} - \frac{2}{\pi} = \frac{5}{3} - \frac{2}{\pi}$$

Q7. Consider the definite integral $I = \displaystyle\int_{-\pi}^{\pi} \left(x^7 e^{x^2} + \frac{\sin^2 x}{1 + 3^x}\right) dx$

Which of the following statement(s) is/are **CORRECT**?

(A) $\displaystyle\int_{-\pi}^{\pi} x^7 e^{x^2} dx = 0$ (B) $\displaystyle\int_{-\pi}^{\pi} \frac{\sin^2 x}{1 + 3^x} dx = \int_{0}^{\pi} \sin^2 x dx$ (C) $\displaystyle\int_{0}^{\pi} \sin^2 x dx = 2\int_{0}^{\pi/2} \sin^2 x dx$

(D) $I = \dfrac{\pi}{2}$

Ans. A, B, C, D

Sol. Let the given integral be split into two parts:

$I = I_1 + I_2$ where $I_1 = \displaystyle\int_{-\pi}^{\pi} x^7 e^{x^2} dx$ and $I_2 = \displaystyle\int_{-\pi}^{\pi} \frac{\sin^2 x}{1 + 3^x} dx$.

1. For I_1:

Let $f(x) = x^7 e^{x^2}$.

Then $f(-x) = (-x)^7 e^{(-x)^2} = -x^7 e^{x^2} = -f(x)$.

Since $f(x)$ is an odd function, by the odd function property:

$$I_1 = \int_{-\pi}^{\pi} x^7 e^{x^2} dx = 0$$

Hence, option (A) is correct.

2. For I_2:

Using the property $\displaystyle\int_a^b g(x)dx = \int_a^b g(a + b - x)dx$ with $a = -\pi$ and $b = \pi$:

$$I_2 = \int_{-\pi}^{\pi} \frac{\sin^2(-x)}{1 + 3^{-x}} dx = \int_{-\pi}^{\pi} \frac{3^x \sin^2 x}{1 + 3^x} dx$$

Adding the two expressions for I_2:

$$2I_2 = \int_{-\pi}^{\pi} \frac{(1 + 3^x)\sin^2 x}{1 + 3^x} dx = \int_{-\pi}^{\pi} \sin^2 x dx$$

Since $h(x) = \sin^2 x$ is an even function, $\displaystyle\int_{-\pi}^{\pi} \sin^2 x dx = 2\int_{0}^{\pi} \sin^2 x dx$.

Thus:

$$2I_2 = 2\int_{0}^{\pi} \sin^2 x dx \implies I_2 = \int_{0}^{\pi} \sin^2 x dx$$

Hence, option (B) is correct.

3. For Queen's Property on $\int_0^{\pi} \sin^2 x dx$:

We have $\int_0^{2a} g(x)dx = 2\int_0^{a} g(x)dx$ if $g(2a-x) = g(x)$.

Here $2a = \pi \implies a = \pi/2$.

Since $\sin^2(\pi - x) = \sin^2 x$, Queen's property gives:

$$\int_0^{\pi} \sin^2 x dx = 2\int_0^{\pi/2} \sin^2 x dx$$

Hence, option (C) is correct.

4. Evaluating the final value:

$I = I_1 + I_2 = 0 + 2\int_0^{\pi/2} \sin^2 x dx = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}$

Hence, option (D) is correct.

Q8. Let $u(x)$ and $v(x)$ be functions defined for all $x > 0$ by

$$u(x) = \int_0^{x} e^{-t^2} dt \quad \text{and} \quad v(x) = \int_0^{x^2} e^{-t} dt$$

If $\frac{d^n u}{dv^n}$ denotes the n-th order derivative of u with respect to v, then which of the following statement(s) is (are) **TRUE**?

(A) $\left.\frac{du}{dv}\right|_{x=1} = \frac{1}{3}$ (B) $\left.\frac{d^2u}{dv^2}\right|_{x=1} = -\frac{e}{4}$ (C) $\left.\frac{d^3u}{dv^3}\right|_{x=1} = \frac{e^2}{8}$

(D) $4x^3\frac{d^2u}{dv^2} + \frac{1}{1-v(x)} = 0$ for all $x > 0$

Ans. B, C, D

Sol. Differentiating $u(x)$ and $v(x)$ with respect to x using Leibniz's Rule gives: $\frac{du}{dx} = e^{-x^2}$

$\frac{dv}{dx} = e^{-x^2} \cdot \frac{d}{dx}(x^2) = 2xe^{-x^2}$

The first derivative of u with respect to v is given by: $\frac{du}{dv} = \frac{\frac{du}{dx}}{\frac{dv}{dx}} = \frac{e^{-x^2}}{2xe^{-x^2}} = \frac{1}{2x}$

At $x = 1$: $\left.\frac{du}{dv}\right|_{x=1} = \frac{1}{2}$

The second derivative of u with respect to v is obtained as: $\frac{d^2u}{dv^2} = \frac{d}{dv}\left(\frac{du}{dv}\right) = \frac{\frac{d}{dx}\left(\frac{1}{2x}\right)}{\frac{dv}{dx}} = \frac{-\frac{1}{2x^2}}{2xe^{-x^2}} = -\frac{e^{x^2}}{4x^3}$

At $x = 1$: $\left.\frac{d^2u}{dv^2}\right|_{x=1} = -\frac{e}{4}$

The third derivative of u with respect to v is: $\frac{d^3u}{dv^3} = \frac{d}{dv}\left(\frac{d^2u}{dv^2}\right) = \frac{\frac{d}{dx}\left(-\frac{1}{4}x^{-3}e^{x^2}\right)}{\frac{dv}{dx}}$

Differentiating the numerator with respect to x:

$\frac{d}{dx}\left(-\frac{1}{4}x^{-3}e^{x^2}\right) = -\frac{1}{4}\left(-3x^{-4}e^{x^2} + 2x^{-2}e^{x^2}\right) = \frac{(3-2x^2)e^{x^2}}{4x^4}$

Dividing by $\frac{dv}{dx} = 2xe^{-x^2}$: $\frac{d^3u}{dv^3} = \frac{(3-2x^2)e^{2x^2}}{8x^5}$

At $x=1$: $\left.\frac{d^3u}{dv^3}\right|_{x=1} = \frac{(3-2)e^2}{8} = \frac{e^2}{8}$

Integrating $v(x)$ directly yields: $v(x) = \left[-e^{-t}\right]_0^{x^2} = 1 - e^{-x^2} \implies 1 - v(x) = e^{-x^2}$

$\implies \frac{1}{1-v(x)} = e^{x^2}$

Substituting $\frac{d^2u}{dv^2} = -\frac{e^{x^2}}{4x^3}$: $4x^3\frac{d^2u}{dv^2} + \frac{1}{1-v(x)} = 4x^3\left(-\frac{e^{x^2}}{4x^3}\right) + e^{x^2} = -e^{x^2} + e^{x^2} = 0$

Paragraph

Consider the parabola $\mathcal{P}: y^2 = 4x$ with focus S. A focal chord PQ of the parabola $\mathcal{P}$ has a length of 16 units, and its slope m is positive. The tangents to $\mathcal{P}$ at points P and Q intersect at point T.

Q9. The area of ΔTPQ is equal to

(A) 16 (B) 24 (C) 32 (D) 48

Ans. C

Sol. For the parabola $\mathcal{P}: y^2 = 4x$, we have $a = 1$

.The length of a focal chord making an angle θ with the x-axis is given by $L = 4a\csc^2\theta$

.Given $L = 16$, we have: $4(1)\csc^2\theta = 16 \implies \csc^2\theta = 4 \implies \sin\theta = \frac{1}{2}$

Since the slope $m > 0$, we get $\theta = \frac{\pi}{6}$ and $m = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$

Let $P = (t_1^2, 2t_1)$ and $Q = (t_2^2, 2t_2)$

Since PQ is a focal chord: $t_1t_2 = -1$

The slope of PQ is given by: $m = \frac{2}{t_1+t_2} = \frac{1}{\sqrt{3}} \implies t_1 + t_2 = 2\sqrt{3}$

Now, $(t_1 - t_2)^2 = (t_1+t_2)^2 - 4t_1t_2$

$= (2\sqrt{3})^2 - 4(-1) = 12 + 4 = 16 \implies |t_1 - t_2| = 4$

The intersection point of the tangents at the ends of a focal chord lies on the directrix $x = -1$: $T = (-a, a(t_1+t_2)) = (-1, 2\sqrt{3})$

The area of ΔTPQ is given by: $\text{Area}(\Delta TPQ) = \frac{1}{2}|t_1 - t_2|(t_1^2 + t_2^2 + 2)$

Since $t_1^2 + t_2^2 = (t_1+t_2)^2 - 2t_1t_2 = 12 + 2 = 14$:

$\text{Area}(\Delta TPQ) = \frac{1}{2} \times 4 \times (14 + 2) = 2 \times 16 = 32$

Q10. The length of the intercept made on the x-axis by the circumcircle of ΔTPQ is :

(A) $2\sqrt{13}$ (B) $4\sqrt{13}$ (C) $8\sqrt{3}$ (D) $4\sqrt{3}$

Ans. B

Sol. Since tangents at the ends of a focal chord intersect at right angles on the directrix, $\angle PTQ = 90^\circ$.

Thus, the circumcircle of ΔTPQ has PQ as its diameter.

The midpoint M of PQ is the center of the circle:

$$M = \left(\frac{t_1^2 + t_2^2}{2}, \frac{2(t_1 + t_2)}{2}\right) = \left(\frac{14}{2}, 2\sqrt{3}\right) = (7, 2\sqrt{3})$$

The radius R of the circle is $\frac{PQ}{2} = \frac{16}{2} = 8$.

The equation of the circumcircle is:

$$(x-7)^2 + (y - 2\sqrt{3})^2 = 8^2 = 64$$

To find the x-intercepts, substitute $y = 0$:

$$(x-7)^2 + (-2\sqrt{3})^2 = 64$$

$$(x-7)^2 + 12 = 64 \implies (x-7)^2 = 52$$

$$\implies x = 7 \pm 2\sqrt{13}$$

The length of the intercept on the x-axis is:
Length $= (7 + 2\sqrt{13}) - (7 - 2\sqrt{13}) = 4\sqrt{13}$

Q1. Let $y(x)$ be a differentiable function defined for $x > -1$ satisfying $y(0) = 0$ and the differential equation

$$(x+1)^2\left(1 + \frac{dy}{dx}\right) = \left(e^{\frac{x+y}{2}} + e^{-\frac{x+y}{2}}\right)^2$$

If $e^{y(1/2)+1/2} = k$, then the value of k is equal to ____.

Ans. 5

Sol. The given differential equation is: $(x+1)^2\left(1 + \frac{dy}{dx}\right) = \left(e^{\frac{x+y}{2}} + e^{-\frac{x+y}{2}}\right)^2$

$$\left(e^{\frac{x+y}{2}} + e^{-\frac{x+y}{2}}\right)^2 = e^{x+y} + 2 + e^{-(x+y)} = \frac{(e^{x+y}+1)^2}{e^{x+y}}$$

Substitute $u = x + y$, which gives $\frac{du}{dx} = 1 + \frac{dy}{dx}$.

The differential equation simplifies to: $(x+1)^2\frac{du}{dx} = \frac{(e^u+1)^2}{e^u}$

$$\frac{e^u}{(e^u+1)^2}du = \frac{dx}{(x+1)^2}$$

$$\int \frac{e^u}{(e^u+1)^2}du = \int \frac{dx}{(x+1)^2}$$

$$-\frac{1}{e^u+1} = -\frac{1}{x+1} + C$$

$$\frac{1}{e^{x+y}+1} = \frac{1}{x+1} + K \quad (\text{where } K = -C)$$

Using the initial condition $y(0) = 0$: $\frac{1}{e^0+1} = \frac{1}{0+1} + K \implies \frac{1}{2} = 1 + K \implies K = -\frac{1}{2}$

Thus, the implicit solution is: $\frac{1}{e^{x+y}+1} = \frac{1}{x+1} - \frac{1}{2} = \frac{1-x}{2(x+1)}$

Now, evaluate at $x = \frac{1}{2}$: $\frac{1}{e^{y(1/2)+1/2}+1} = \frac{1-1/2}{2(1+1/2)} = \frac{1/2}{3} = \frac{1}{6}$

$e^{y(1/2)+1/2} + 1 = 6 \implies e^{y(1/2)+1/2} = 5$

Q2. Consider the ellipse $E : \frac{x^2}{2} + y^2 = 1$. A family of parallel chords of E has a constant slope $m = 1$. Let $y = x + c$ be the line containing any such chord, where $c \in [-\sqrt{3}, \sqrt{3}]$. Let $L(c)$ denote the length of the chord of the ellipse E intercepted by the line $y = x + c$.

If $I = \int_0^{\sqrt{3}} c \cdot [L(c)]^2 \, dc$ then the value of I is _____.

Ans. 4

Sol. Substitute $y = x + c$ into the equation of the ellipse $x^2 + 2y^2 = 2$:

$\implies 3x^2 + 4cx + (2c^2 - 2) = 0$

Let x_1 and x_2 be the x-coordinates of the points of intersection of the line and the ellipse.

$x_1 + x_2 = -\frac{4c}{3}, \quad x_1 x_2 = \frac{2c^2-2}{3}$

The square of the difference of the roots is given by:

$(x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2 = \frac{16c^2}{9} - 4\left(\frac{2c^2-2}{3}\right) = \frac{24 - 8c^2}{9} = \frac{8(3-c^2)}{9}$

Since the slope of the line is $m = 1$, the length of the chord $L(c)$ is:

$L(c) = \sqrt{(1+m^2)(x_1-x_2)^2} = \sqrt{2 \cdot \frac{8(3-c^2)}{9}} = \frac{4}{3}\sqrt{3-c^2}$

Taking the square of $L(c)$: $[L(c)]^2 = \frac{16}{9}(3-c^2)$

$$I = \int_0^{\sqrt{3}} c \cdot \frac{16}{9}(3-c^2)\, dc = \frac{16}{9}\int_0^{\sqrt{3}} (3c - c^3)\, dc$$

Thus, $I = \frac{16}{9} \times \frac{9}{4} = 4$

Q3. Let A be the area of the region bounded by the curves $y = 3x^2 - 3$ and $y = 3 - 3|x|$. Find the value of A.

Ans. 7

Sol. The given bounded region is enclosed by the curves $y_1(x) = 3 - 3|x|$ and $y_2(x) = 3x^2 - 3$.

To determine the points of intersection, consider $x \geq 0$:

$3 - 3x = 3x^2 - 3 \implies 3x^2 + 3x - 6 = 0$

$\implies x^2 + x - 2 = 0$

Solving this gives $x = 1$ (since $x \geq 0$).

By symmetry about the y-axis, the points of intersection are $x = -1$ and $x = 1$.

For $x \in [-1, 1]$, we have $y_1(x) \geq y_2(x)$.

Using the symmetry of both functions, the area A is:

$A = 2\displaystyle\int_0^1 \big((3 - 3x) - (3x^2 - 3)\big)dx$

$A = 2\displaystyle\int_0^1 (6 - 3x - 3x^2)dx = 6\int_0^1 (2 - x - x^2)dx$

$A = 6\left[2x - \dfrac{x^2}{2} - \dfrac{x^3}{3}\right]_0^1 = 6\left(2 - \dfrac{1}{2} - \dfrac{1}{3}\right) = 6\left(\dfrac{7}{6}\right) = 7$

Q4. Let $f : [0, \infty) \to \mathbb{R}$ be a function defined by

$f(x) = (x + 1)\ln(1 + x) - x - \dfrac{x^2}{k + x}$

where k is a positive real parameter. Let S be the set of all values of $k > 0$ for which the inequality $f(x) \geq 0$ holds for all $x \geq 0$. If $m = \min S$, then the value of m is equal to

Ans. 2

Sol. Given $f(x) = (x + 1)\ln(1 + x) - x - \dfrac{x^2}{k + x}$ for $x \geq 0$ and $k > 0$. Note that $f(0) = 0$.

Differentiating $f(x)$ with respect to x:

$f'(x) = \ln(1 + x) + (x + 1) \cdot \dfrac{1}{1 + x} - 1 - \dfrac{2x(k + x) - x^2}{(k + x)^2} = \ln(1 + x) - \dfrac{2kx + x^2}{(k + x)^2}$

Notice that $f'(0) = 0$. Rewrite $f'(x)$ as: $f'(x) = \ln(1 + x) - \left(1 - \dfrac{k^2}{(k + x)^2}\right)$

Differentiating $f'(x)$ with respect to x: $f''(x) = \dfrac{1}{1 + x} - \dfrac{2k^2}{(k + x)^3} = \dfrac{(k + x)^3 - 2k^2(1 + x)}{(1 + x)(k + x)^3}$

Let $P(x) = (k + x)^3 - 2k^2(1 + x) = x^3 + 3kx^2 + k^2x + k^2(k - 2)$.

Case 1: $k \geq 2$

Since $k > 0$ and $k - 2 \geq 0$, we have $P(x) > 0$ for all $x > 0$.

Thus, $f''(x) > 0$ for all $x > 0$, which implies $f'(x)$ is strictly increasing on $[0, \infty)$.

Since $f'(0) = 0$, $f'(x) > 0$ for all $x > 0$.

Consequently, $f(x)$ is strictly increasing on $[0, \infty)$, yielding $f(x) > f(0) = 0$ for all $x > 0$.

Therefore, $f(x) \geq 0$ holds for all $x \geq 0$.

Case 2: $0 < k < 2$

Here $P(0) = k^2(k-2) < 0$.

By continuity of $P(x)$, there exists some $\delta > 0$ such that $P(x) < 0$ for all $x \in [0, \delta)$.

This implies $f''(x) < 0$ on $(0, \delta)$, so $f'(x)$ is strictly decreasing on $[0, \delta)$. Since $f'(0) = 0$, $f'(x) < 0$ for $x \in (0, \delta)$, making $f(x)$ strictly decreasing on $[0, \delta)$.

Thus, $f(x) < f(0) = 0$ for $x \in (0, \delta)$, which violates the inequality.

Hence, $S = [2, \infty)$, giving $m = \inf S = 2$.

Q1. Let $f : [0,1] \to [0, \infty)$ be a continuous function such that $f(0) = 0$ and $f(x) > 0$ for all $x \in (0,1]$. Suppose $f(x)$ satisfies the integral equation

$$f(x) \int_0^x \frac{f(t)}{\sqrt{1-t^2}}\, dt = \frac{x^3\sqrt{1-x^2}}{2} \quad \text{for all } x \in (0,1)$$

Let M be the maximum value of $f(x)$ on the interval $[0,1]$, attained at $x = x_0$.

Let A_1 be the area of the region bounded by the curve $y = f(x)$, the horizontal line $y = M$, and the y-axis for $x \in [0, x_0]$.

Let A_2 be the area of the region bounded by the curve $y = f(x)$, the x-axis, and the vertical line $x = x_0$ for $x \in [0, x_0]$.

Find the value of $3A_1 + 12A_2$.

Ans. 3

Sol. Let $g(x) = \displaystyle\int_0^x \frac{f(t)}{\sqrt{1-t^2}}\, dt$.

$$g'(x) = \frac{f(x)}{\sqrt{1-x^2}} \implies f(x) = g'(x)\sqrt{1-x^2}$$

Substituting $f(x)$ into the given integral equation yields

$$g'(x)\sqrt{1-x^2} \cdot g(x) = \frac{x^3\sqrt{1-x^2}}{2}$$

$$g'(x)g(x) = \frac{x^3}{2} \implies \frac{1}{2}\frac{d}{dx}\left[(g(x))^2\right] = \frac{x^3}{2}$$

Integrating both sides with respect to x from 0 to x (noting $g(0) = 0$):

$$(g(x))^2 = \frac{x^4}{4} \implies g(x) = \frac{x^2}{2} \quad (\text{since } g(x) > 0 \text{ for } x > 0)$$

Thus, $f(x) = g'(x)\sqrt{1-x^2} = x\sqrt{1-x^2}$

To find the maximum of $f(x)$ on $[0,1]$: $f'(x) = \sqrt{1-x^2} - \dfrac{x^2}{\sqrt{1-x^2}} = \dfrac{1-2x^2}{\sqrt{1-x^2}}$

Setting $f'(x) = 0$ gives $x_0 = \dfrac{1}{\sqrt{2}}$. The maximum value is $M = f\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{1}{2}$.

Now, compute A_2: $A_2 = \displaystyle\int_0^{1/\sqrt{2}} x\sqrt{1-x^2}\, dx = \left[-\frac{1}{3}(1-x^2)^{3/2}\right]_0^{1/\sqrt{2}}$

$= \dfrac{1}{3}\left(1 - \dfrac{1}{2\sqrt{2}}\right) = \dfrac{4-\sqrt{2}}{12}$

The region corresponding to $A_1 + A_2$ forms a rectangle with width $x_0 = \frac{1}{\sqrt{2}}$ and height $M = \frac{1}{2}$.

Area of rectangle $= A_1 + A_2 = \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{2}}{4}$

Thus, $A_1 = \frac{\sqrt{2}}{4} - A_2 = \frac{\sqrt{2}}{4} - \frac{4-\sqrt{2}}{12} = \frac{4\sqrt{2}-4}{12} = \frac{\sqrt{2}-1}{3}$

Finally, evaluate $3A_1 + 12A_2$: $3A_1 + 12A_2 = 3\left(\frac{\sqrt{2}-1}{3}\right) + 12\left(\frac{4-\sqrt{2}}{12}\right)$

$= (\sqrt{2}-1) + (4-\sqrt{2}) = 3$

Q2. Evaluate the value of the following limit:

$L = \lim\limits_{n\to\infty} \frac{1}{n} \sum\limits_{r=1}^{8n} \left(\sin^4\left(\frac{\pi r}{n}\right) + \cos^4\left(\frac{\pi r}{n}\right)\right)$

Ans. 6

Sol. Expressing the given limit of sum as a definite integral: $L = \int_0^8 \left(\sin^4(\pi x) + \cos^4(\pi x)\right) dx$

Let $f(x) = \sin^4(\pi x) + \cos^4(\pi x)$.

Hence, $f(x)$ is a periodic function with period $T = \frac{1}{2}$.

$L = \int_0^{16\times\frac{1}{2}} f(x)dx = 16\int_0^{1/2} \left(\sin^4(\pi x) + \cos^4(\pi x)\right) dx$

Substitute $\pi x = t \implies dx = \frac{dt}{\pi}$: $\int_0^{1/2} \left(\sin^4(\pi x) + \cos^4(\pi x)\right) dx = \frac{1}{\pi}\int_0^{\pi/2} (\sin^4 t + \cos^4 t) dt$

Using Walli's formula: $\int_0^{\pi/2} \sin^4 t\, dt = \int_0^{\pi/2} \cos^4 t\, dt = \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{16}$

So, $\int_0^{1/2} f(x)dx = \frac{1}{\pi}\left(\frac{3\pi}{16} + \frac{3\pi}{16}\right) = \frac{3}{8}$

Thus, $L = 16 \times \frac{3}{8} = 6$

Q3. Let $f(x)$ and $g(x)$ be functions such that

$\int e^x \left(\frac{1-\sin x}{1-\cos x}\right) dx = -e^x f(x) + C_1$ and $\int \left(\frac{x+\sin x}{1+\cos x}\right) dx = g(x) + C_2$

where C_1 and C_2 are constants of integration. If $f\left(\frac{\pi}{2}\right) + \frac{2}{\pi} g\left(\frac{\pi}{2}\right) = k$, then find the value of k.

Ans. 2

Sol. For the first integral, rewriting the trigonometric terms in terms of half angles:

$\int e^x \left(\frac{1-\sin x}{1-\cos x}\right) dx = \int e^x \left(\frac{1 - 2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\sin^2\left(\frac{x}{2}\right)}\right) dx$

$= \int e^x \left(-\cot\left(\frac{x}{2}\right) + \frac{1}{2}\csc^2\left(\frac{x}{2}\right)\right) dx$

Using the classic integral formula $\int e^x(h(x)+h'(x))dx = e^x h(x) + C$:

Here, $h(x) = -\cot\left(\frac{x}{2}\right)$, so $h'(x) = \frac{1}{2}\csc^2\left(\frac{x}{2}\right)$.

Thus,

$$\int e^x \left(\frac{1-\sin x}{1-\cos x}\right)dx = -e^x \cot\left(\frac{x}{2}\right) + C_1$$

Comparing with the given expression gives $f(x) = \cot\left(\frac{x}{2}\right)$.

Therefore, $f\left(\frac{\pi}{2}\right) = \cot\left(\frac{\pi}{4}\right) = 1$.

For the second integral:

$$\int \left(\frac{x+\sin x}{1+\cos x}\right)dx = \int \left(\frac{x + 2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\left(\frac{x}{2}\right)}\right)dx$$

$$= \int \left(\tan\left(\frac{x}{2}\right) + x \cdot \frac{1}{2}\sec^2\left(\frac{x}{2}\right)\right)dx$$

Using the classic integral formula $\int (u(x) + xu'(x))dx = xu(x) + C$:

Here, $u(x) = \tan\left(\frac{x}{2}\right)$, so $u'(x) = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)$.

Thus,

$$\int \left(\frac{x+\sin x}{1+\cos x}\right)dx = x\tan\left(\frac{x}{2}\right) + C_2$$

Comparing gives $g(x) = x\tan\left(\frac{x}{2}\right)$.

Therefore, $g\left(\frac{\pi}{2}\right) = \frac{\pi}{2}\tan\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$.

Substituting these values to find k:

$$k = f\left(\frac{\pi}{2}\right) + \frac{2}{\pi}g\left(\frac{\pi}{2}\right)$$

$$= 1 + \frac{2}{\pi}\left(\frac{\pi}{2}\right) = 1 + 1 = 2$$